

Dipayan Biswas

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Summary

Machine learning researcher and Ph.D. candidate (expected Dec 2026) specializing in multimodal AI: vision-language models, LLM evaluation, and computer vision. Deployed a summarization system used by 750+ students, fine-tuned VLMs with QLoRA, and published 5 peer-reviewed IEEE papers.

Education

University of Houston

Ph.D. in Computer Science, GPA 3.88

M.Sc. in Computer Science (awarded en route to the Ph.D.)

Houston, TX

Aug 2021 – Dec 2026 (Expected)

May 2026

Khulna University of Engineering & Technology (KUET)

B.Sc. in Electronics and Communication Engineering, GPA 3.88

Khulna, Bangladesh

Apr 2012 – May 2016

Technical Skills

AI/ML: Multimodal AI, Vision-Language Models (VLMs), Large Language Models (LLMs), LLM Evaluation (LLM-as-judge), Prompt Engineering, Fine-tuning (LoRA, QLoRA), Computer Vision, Object Detection

Frameworks: PyTorch, Hugging Face (Transformers, PEFT, TRL), OpenCV, LiteLLM, LangChain, DeepEval, scikit-learn

Models & APIs: Gemini, OpenAI, Anthropic, Gemma 3, Qwen3-VL, SigLIP, CLIP, YOLO, Faster R-CNN, DETR

Programming & MLOps: Python, SQL, C++, Git, FastAPI, vLLM, Ollama, MLflow, Slurm, AWS

Experience

University of Houston

Graduate Research / Teaching Assistant

Houston, TX

Aug 2021 – Present

Multimodal Video Summarization (NSF-funded VideoPoints platform)

- Condensed hour-long lectures into 5-minute summaries with supporting visuals for review and navigation, used by 750+ students in 10+ courses; 78% of surveyed students rated them helpful.
- Designed a multimodal pipeline fusing speech (Whisper), slide text (OCR), and slides with Gemma 3, raising faithfulness by 19% (76.7% to 91.4%) and coverage by 15% in a blind evaluation by 8 faculty.
- Automated evaluation with an LLM-as-judge (DeepEval) that matched faculty agreement on faithfulness (0.68 vs. 0.59 rank correlation) and reproduced their 19% gain (judge: 18%).
- Showed that per-slide fusion lifts the coverage gain over transcript-only from 10.6% to 15.0%, beyond run-to-run noise.
- Developed zero-shot cross-modal selection of supporting visuals with SigLIP and Gemma 3; faculty rated it 82.8% (up from 70.3%) and kept 83% of the selected visuals.

Object Detection in Presentation Videos

- Released LVVO, a 4,000-frame dataset from 245 videos, planning the labeling guidelines and a multi-annotator protocol with expert conflict resolution (83.4% inter-annotator F1).
- Raised AP50 from 90.8% to 95.3% with semi-supervised auto-labeling, adding 3x more training frames to 1,000 expert-labeled frames without new manual labeling.
- Developed a graph-based method merging fragmented detections into coherent visuals, improving mAP by 15.8%.

Fellowship.AI

Machine Learning Fellow

Remote

Sep 2025 – Nov 2025

- Developed a multimodal listing analyzer (Gemini API) that jointly scores real estate photos and descriptions to recommend pricing, staging, and content improvements across 3k listings.
- Engineered a weighted-KNN price prediction pipeline with outlier detection over 8K+ Redfin listings, flagging mispriced properties at 85% precision against held-out sale prices.

Khulna University of Engineering & Technology (KUET)

Assistant Professor / Lecturer

Khulna, Bangladesh

Feb 2017 – Aug 2021

- Taught five undergraduate courses with labs, including Digital Image Processing, and supervised student research projects.
- Mentored 10+ undergraduates in ML research, co-authoring IEEE papers on plant disease (99.80% accuracy) and breast cancer (99.28%) classification with transfer learning and ensemble methods.

Samsung R&D Institute Bangladesh (SRBD)

Software Engineer

Dhaka, Bangladesh

Jul 2016 – Feb 2017

- Implemented 360-degree photo and video viewing modes, including panoramic and little-planet views, for the shipped Samsung Gear 360 iOS app using Objective-C, OpenGL ES, and OpenCV.

Projects

Unified Inference Toolkit for Vision-Language Models [\[Code\]](#)

Jun 2026 – Aug 2026

- Built one interface for multimodal prompts on 8+ backends, from cloud APIs to vLLM, Ollama, and Hugging Face Transformers.
- Pre-configured 10+ open LLMs and VLMs, including Gemma 3, Qwen3-VL, and InternVL, each callable by name.

Multi-Step Workflow Framework for LLMs and VLMs [\[Code\]](#)

Jun 2026 – Sep 2026

- Built configurable prompt chaining for multi-step workflows on local or cloud models, with resumable experiments.
- Made the workflow engine reusable across datasets, saving every step's output for review.

VLM Fine-tuning for Visual Question Answering [\[Code\]](#)

May 2026 – Jun 2026

- Fine-tuned Gemma 3 with QLoRA, raising ChartQA accuracy from 43.6% to 61.6% (4B) and 61.8% to 69.4% (12B).
- Trained on 28k chart questions using Hugging Face TRL and PEFT on single and multiple L40S GPUs via Slurm.

Lecture Video Chaptering with LLMs [\[Details\]](#)

Jun 2025 – Aug 2025

- Deployed automatic chaptering that splits each newly uploaded lecture into titled chapters; in daily use since Aug 2025.
- Aligned Whisper transcripts with slide transitions and used Gemini structured output, auto-correcting gaps and overlaps.

Modular Object Detection Training Pipeline [\[Code\]](#)

Dec 2024 – Feb 2025

- Designed a PyTorch codebase for reproducible training and fine-tuning of YOLO, Faster R-CNN, and DETR, with Hydra configs and unified train, evaluate, and predict steps.

Small and Distant Face Detection [\[Code\]](#)

Feb 2023 – May 2023

- Trained and evaluated an IoU-aware RetinaFace with ResNet50 and MobileNet0.25 backbones on 4 V100 GPUs, raising AP on the medium and hard WIDER FACE sets.

Publications

- **D. Biswas**, S. Shah, and J. Subhlok. “Visual Content Detection in Educational Videos with Transfer Learning and Dataset Enrichment.” *IEEE MIPR 2025*. [\[Paper\]](#) [\[Code\]](#)
- **D. Biswas**, S. Shah, and J. Subhlok. “Lecture Video Visual Objects (LVVO) Dataset: A Benchmark for Visual Object Detection in Educational Videos.” *arXiv 2025*. [\[Paper\]](#) [\[Data\]](#)
- **D. Biswas**, S. Shah, and J. Subhlok. “Identification of Visual Objects in Lecture Videos with Color and Keypoints Analysis.” *IEEE ISM 2023*. [\[Paper\]](#)
- I. Z. Mukti and **D. Biswas**. “Transfer Learning Based Plant Diseases Detection Using ResNet50.” *IEEE EICT 2019*. [\[Paper\]](#)
- S. Das and **D. Biswas**. “Prediction of Breast Cancer Using Ensemble Learning.” *IEEE ICAEE 2019*. [\[Paper\]](#)
- **D. Biswas** and M. F. Hossain. “Epileptic Seizure Detection Based on Selected Features of Different Complexities Using ANN.” *IEEE EICT 2017*. [\[Paper\]](#)

Honors & Awards

- Ph.D. Research Showcase, Audience's Choice Runner-Up, University of Houston, 2025
- Cullen Graduate Student Fellowship, University of Houston, 2021–2026
- University Gold Medalist (ranked 1st in class), KUET, 2018

Certifications & Courses

- Generative AI with Large Language Models, DeepLearning.AI and AWS [\[Certificate\]](#)
- Machine Learning in Production, DeepLearning.AI [\[Certificate\]](#)
- Building and Evaluating Advanced RAG, DeepLearning.AI [\[Course\]](#)
- AWS Cloud Practitioner Essentials, Amazon Web Services (course) [\[Certificate\]](#)

Leadership & Service

- Organizing Secretary, Graduate Student Association of Bangladesh, University of Houston, 2022–2025
- Academic Advisor and Project Mentor, KUET, 2017–2021